

SOORYA PRAKASH S

Game & XR Developer

Trichy, Tamil Nadu, India | sooryapraakash71@gmail.com

linkedin.com/in/soorya-prakash-s-486733128 | sooryapraakash71.github.io/portfolio

SUMMARY

Game and XR developer working in Unity and C#, building experiences across 2D, 3D, VR, AR and Apple Vision Pro. At Tata Consultancy Services since 2023, building gameplay and interaction systems, UI flows and spatial experiences. Outside work, builds games from the ground up, including custom physics, level solvers and multiplayer networking without third-party libraries.

EXPERIENCE

Developer (XR & Unity), Tata Consultancy Services (TCS), Chennai Jul 2023 - Present

- Develop Unity-based 2D, 3D and XR applications in C#, and handle debugging, performance optimization and multi-platform builds for Android, Windows and Apple Vision Pro.
- Built a VR training simulation with physical grab and manipulate interactions, teleport and smooth locomotion, step-by-step training tasks, and visual, audio and haptic feedback (XR Interaction Toolkit, OpenXR, Oculus SDK, AutoHands).
- Built an Apple Vision Pro prototype with Unity, PolySpatial and visionOS, using gaze targeting and pinch selection for controller-free input and spatial UI.
- Built an AR prototype with AR Foundation and ARCore covering surface detection, object placement, and touch and gesture controls to move, rotate and scale objects.
- Built a 3D interactive application with player movement, physics, camera systems, UI and input handling.

PROJECTS

Gin Distillation Training Simulation VR | *Unity, C#, XR Interaction Toolkit, OpenXR*

VR simulation of a 16-step gin distillation startup procedure: interactive valves, pump control panel, still hatch and aromatics, with temperature, spirit level and foam monitoring driven by procedural state logic.

Gravity Drift | *Unity 6, C#, 2D Physics, Procedural Graphics, Runtime Audio*

2D physics platformer with six levels and a tutorial. The player is a Rigidbody2D moved by force and torque, with per-player gravity flip zones, wind, a HingeJoint2D pendulum, and core visuals and several sound effects generated at runtime.

Glowworm | *Unity 6, C#, Procedural Generation*

Children's learning game where a worm eats the next item in a sequence, so one mechanic covers counting, letters and spelling, with generated levels checked by a solver.

GridPush | *Unity 6, C#*

15-level Sokoban game with a Unity-independent rules engine and a breadth-first search solver that verifies each level and orders levels by optimal solution length.

Chess | *Unity, C#, TCP, UDP, WebSocket*

One rules engine behind three modes: pass and play, LAN over TCP with UDP discovery, and online through a custom WebSocket relay server, with no Photon, Mirror or networking library. About 5,900 lines of C#, 81 automated tests, move generation verified with perft.

Hotspot Founder | *Unity 6, C#, uGUI*

Image-based learning tool: load any image at runtime, place and label hotspots, then explore them by hovering. Positions are stored relative to the image, so content works at any size or aspect ratio.

VR Bowling | *Unity, C#, XR Interaction Toolkit, OpenXR, Meta Quest*

Physics-driven bowling for Meta Quest 2 and 3 with controller and hand-tracking throws, pin physics and standard scoring.

More projects and gameplay videos: sooryapraakash71.github.io/portfolio

SKILLS

Game Development: Unity 2D & 3D, Gameplay Programming, Game Architecture, Physics, Animation, UI (uGUI), Game State Management, Procedural Generation, Editor Tooling

XR & Spatial: XR Interaction Toolkit, OpenXR, AR Foundation, ARCore, ARKit, Meta Quest SDK, PolySpatial, visionOS, Hand Tracking, Gaze Tracking, Spatial UI

Programming: C#, Object-Oriented Programming, SOLID Principles, Design Patterns, Data Structures & Algorithms, State Machines, Event-Driven Systems, Debugging, Code Optimization

Networking: Netcode for GameObjects, TCP, UDP, WebSocket, Client/Server Architecture, Relay Systems, Lag Compensation

Tools: Git, GitHub, Unity Profiler, Unity Package Manager, Addressables, Assembly Definitions, ScriptableObjects

Platforms: Windows, Android, iOS, WebGL, Meta Quest, Apple Vision Pro, macOS

EDUCATION

Master of Computer Applications (MCA), SASTRA Deemed University, CGPA 8.71

2023 - 2025

Bachelor of Science in Computer Science, Bishop Heber College, CGPA 8.13

2020 - 2023